

CHAPTER 2

LITERATURE REVIEW

2.1. Jet Engine Principles and Mechanics

A GTE, commonly known as jet engine, is a heat engine that uses air as a work fluid, as means to provide thrust. This principle is based on Sir Isaac Newton's theory, that for every action, there is an equal and opposite reaction. All jet engines work on the same principle. The air is firstly sucked in the front of the engine, by a fan. This air then passes through a compressor, which has the function to raise the pressure of the air to enhance the combustion. The compressed air is then injected into the combustor along with fuel, which is injected through fuel injectors, and then a sparkplug lights the mixture. Here the mixture's temperature rises, and the burning gases expand and gain velocity through the nozzle, but in the way they pass through a set of blades, that are called the turbine, making them spin, which in turn produces power. The turbine is attached to the same shaft as the compressor, and the power output of the turbine drives the compressor. By the time the hot gas reaches the turbine, its pressure is still above that of the surroundings, and the final expansion takes place in the nozzle, in which the exhaust gas velocity increases. It is then this high-velocity jet that produces the thrust to propel the aircraft through the air.

2.2. The Working Cycle

The working cycle of a gas turbine engine (GTE) resembles that of an internal combustion engine (ICE) in that both involve intake, compression, combustion, and exhaust processes. However, key differences exist between the two. In an ICE, combustion occurs at constant volume, and the processes are intermittent, occurring in distinct stages within the engine's cylinders.

Conversely, in a GTE, combustion takes place at constant pressure, and the processes are continuous. Air is continuously compressed, mixed with fuel, ignited, and then expelled as high-energy gases that drive the turbine. This continuous operation aligns with the principles of the Brayton cycle, which defines the thermodynamic path of gas turbines, enabling more efficient energy conversion for applications like aviation.

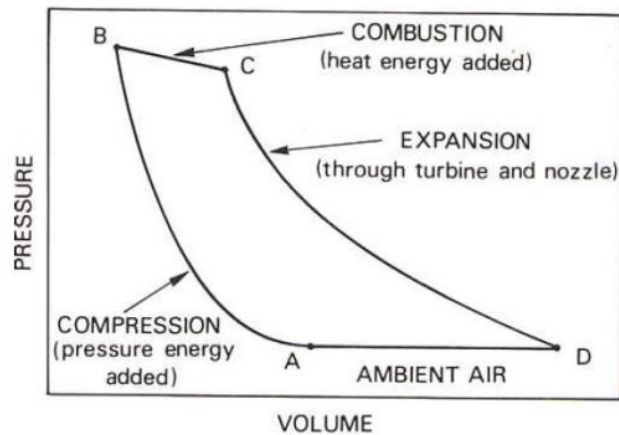


Figure 2.1. Pressure-Volume Diagram of the Brayton Cycle

- i. A-B – Point A represents air at atmospheric pressure, which is then compressed along the line AB, denoting the compressor cycle. This compression raises the air's temperature, so the air exits the compressor and enters the combustor at higher pressure and temperature. The pressure ratio, determined by the compressor's performance, is a key factor in the engine's overall thermodynamic efficiency. A higher pressure ratio leads to greater thermodynamic efficiency.
- ii. B-C – Heat is added from B to C by introducing fuel at constant pressure, significantly increasing the air volume. The fuel, typically natural or petroleum products, provides the energy needed to drive the cycle. The pressure drop between these points represents the pressure drop in the combustion chambers, which should be kept very low.
- iii. C-D – From C to D, the combustion gases expand through the turbine and jet pipe to the atmosphere. During this phase, the turbine converts some of the energy in the expanding gases into mechanical power, while the remaining gases produce a propulsive jet.

The expansion of gases in a gas turbine engine increases efficiency and power output by maximizing thermal energy. This process allows the engine to convert high-pressure, high-temperature gases into mechanical energy, which drives the turbine and other engine components. Advanced cooling techniques, including internal air channels and thermal barrier coatings, are used to protect turbine components and maintain engine longevity. These innovations improve heat resistance, enhancing thermal management and contributing to the overall durability and performance of the engine. Effective cooling ensures that turbines operate safely even under extreme conditions.

2.3. The Compressors

The compressor is the first component of the engine core. It consists of fans attached to a shaft, with blades designed to squeeze the air into progressively smaller areas, increasing its pressure. Today, most jet engines use axial-flow compressors where air flows in one direction along the shaft connecting the compressor and turbine. The air moves through alternating rows of stationary and rotating blades, known as stators and rotors, which decrease velocity and increase pressure. Early commercial jet engines had single shafts, limiting airflow matching, but the introduction of a second shaft allowed for twin-spool engines, as shown in Figure 2.2. This design improves efficiency by matching compression stages more accurately to airflow.

In twin-spool engines, the Low-Pressure Compressor (LPC) and High-Pressure Compressor (HPC) operate at different speeds, optimizing airflow management for better performance. Twin-spool compressors are also used in bypass engines, where the LPC handles larger airflow than the HPC. The bypass ratio, characteristic of turbofan engines, compares air exiting without going through the core (secondary flow) to air going through the core (primary flow). Most thrust in turbofan engines comes from the fan, part of the LPC (Figure 2.2), driven by the Low-Pressure Turbine (LPT). Turbofan engines are quieter and more fuel-efficient, with high bypass ratios making the fan stage responsible for 75-85% of the total thrust.

Additionally, advancements in blade aerodynamics and materials allow compressors to achieve higher pressure ratios without compromising engine durability. These innovations ensure that axial-flow compressors continue to evolve, driving further improvements in engine performance and environmental compliance.

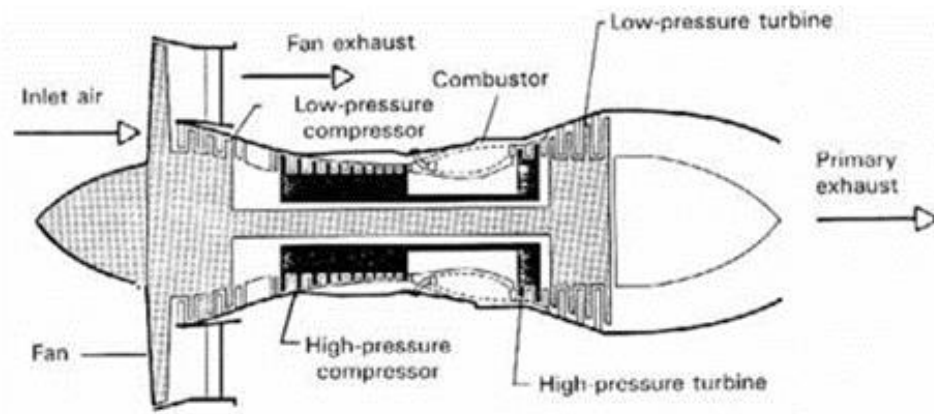


Figure 2.2. A Cut View of a Twin-Spool Turbofan Engine

2.4. The Turbines

The turbine's function is to drive the compressor and accessories, and in some applications, to provide shaft power for a propeller or rotor. Like the compressor, the turbine has stages with stationary nozzle guide vanes and moving blades. These stages are categorized into High-Pressure Turbine (HPT) and Low-Pressure Turbine (LPT), each with its own shaft.

The HPT, closest to the combustor, deals with high-temperature gases. Combustion engineers and HPT engineers collaborate to address challenges in this section. Hot gases accelerate through convergent nozzles, spinning the flow and causing the turbine to rotate at high speeds, providing power for the turbine shaft and compressor. Turbines operate under high stresses, with temperatures between 850 and 1700°C and speeds up to 762 m/s. The inlet temperature is defined by the combustion process, which should ideally be as high as possible for optimal performance. Despite advances in nickel alloys, turbine blades cannot fully withstand these temperatures, so cooling techniques have been developed. These include blade film cooling, where air from the HPC passes through internal channels and ejects at the blade edges, and ceramic coatings to enhance heat resistance and reduce cooling air requirements, improving engine efficiency.

2.5. The Nozzle

The nozzle is the exhaust duct of the engine, and has an important role in providing thrust for the aircraft. The energy of the airflow that passed the turbine stages, in addition to the cooler air that bypassed the engine core, meet at the exit of the nozzle and produce a force that acts to propel the engine forward, which is called thrust. In order to make the engine quieter, a mixer is implemented before the nozzle, to combine the high temperature air coming from the engine core, with the lower temperature air that bypasses in the fan.

2.6. Types of Aircraft Engines

1. Reciprocating Internal Combustion Engine
2. Turbo-Prop Engine
3. Turbojet Engine
4. Turbofan Engine
5. Ramjet Engine

2.7. Reciprocating Internal Combustion Engine

An aircraft piston engine, also commonly referred to as a reciprocating engine or "recip", is an internal combustion engine that uses one or more reciprocating pistons to convert pressure into a rotational motion. The aircraft piston engine operates on the same principles as the engines found in most automobiles. However, modifications, such as dual ignition systems, to improve redundancy and safety, and air cooling to reduce weight, have been incorporated into engines designed for aviation use. Turbochargers and, less commonly, superchargers can be added to piston engines to improve performance. Aircraft piston engines are most commonly fueled with AVGAS but diesel fueled engines are becoming more common, especially in light aircraft.

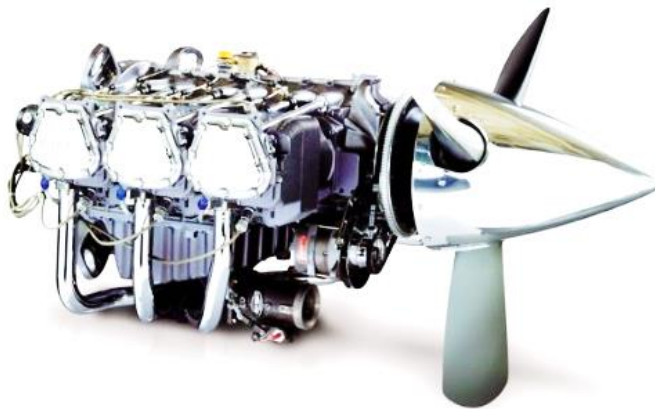


Figure 2.3. Reciprocating Internal Combustion Engine

2.8. Turbo-Prop Engine

A turboprop is a type of turbine engine designed to drive an aircraft propeller, integrating jet propulsion principles with traditional propeller-driven technology. The engine consists of several key components: an intake, reduction gearbox, compressor, combustor, turbine, and propelling nozzle. Air enters the engine through the intake and is compressed by the compressor, which increases its pressure and temperature. The compressed air then flows into the combustor, where fuel is added and ignited, resulting in high-temperature and high-pressure gases. These gases expand through the turbine stages, generating power, some of which drives the compressor and an electric generator. The remaining exhaust gases are expelled from the turbine, contributing to the aircraft's propulsion. This configuration allows the turboprop engine to effectively combine the efficiency of a propeller with the power of jet propulsion.



Figure 2.4. Turbo-Prop Engine

2.9. Turbojet Engine

The turbojet is an airbreathing jet engine typically used in aircraft. It consists of a gas turbine with a propelling nozzle. The gas turbine has an air inlet that includes inlet guide vanes, a compressor, a combustion chamber, and a turbine that drives the compressor. The compressed air from the compressor is heated by burning fuel in the combustion chamber and then allowed to expand through the turbine. The turbine exhaust is then expanded in the propelling nozzle, where it is accelerated to high speed to provide thrust. Turbojets operate efficiently at high speeds, making them ideal for supersonic aircraft. However, they tend to be less fuel-efficient at subsonic speeds compared to turbofans, which are more common in modern commercial aviation. Despite this, turbojets are still favored in military and high-speed applications where maximum speed and performance take precedence over fuel economy.



Figure 2.5. Turbojet Engine

2.10. Turbofan Engine

The turbofan or fanjet is a type of airbreathing jet engine that is widely used in aircraft propulsion. It consists of a gas turbine engine which achieves mechanical energy from combustion, and a ducted fan that uses the mechanical energy from the gas turbine to force air rearwards. Thus, whereas all the air taken in by a turbojet passes through the combustion chamber and turbines, in a turbofan some of that air bypasses these components. A turbofan thus can be thought of as a turbojet being used to drive a ducted fan, with both of these contributing to the thrust.



Figure 2.6. Turbofan Engine

2.11. Ramjet Engine

A ramjet is a variant of an air breathing jet engine that does not include a rotary compressor; rather, it uses the engine's forward motion to compress the incoming air. A ramjet cannot function at zero airspeed and therefore cannot be used to power an aircraft in all phases of flight. A ramjet equipped aircraft requires another type of propulsion to accelerate it to a speed at which the ramjet is capable of producing thrust.

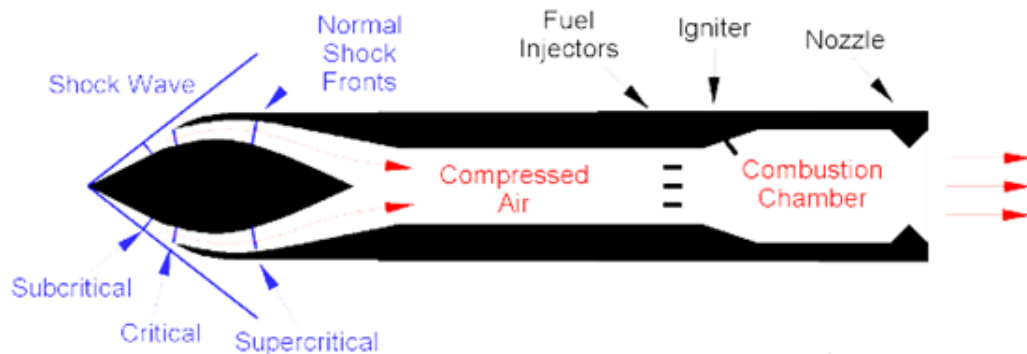


Figure 2.7. Ramjet Engine

2.12. The Combustor

The goal of the combustor is to convert the chemical energy bound in the fuel into thermal energy. This thermal energy, is the energy used by the turbine to produce the power required to operate the various stages of compressors, or in the case of an industrial GTE, the turbine produces the power required to turn a generator, which in turn produces electricity. It is important to understand the difference between the combustor and the Combustion Chamber (CC). The combustor includes all of the combustion systems, i.e. the combustor is the conjunction of the diffuser, the combustion chamber, the inner and outer casings, the spark plugs and the fuel injectors, whereas the CC refers to the exact place in which combustion takes place.

The combustor is a critical component in the GTE, because it must operate reliably at extreme temperatures, it has to provide a suitable temperature for the turbine inlet, and it must produce a minimum amount of pollutants over a long operating life. All of the challenges of these aspects, along with combustor fundamentals, will be explained with detail in this section.

2.13. Combustion Process

The combustion of a given liquid fuel involves the mixing of a fine spray of droplets with air, the vaporization of these droplets, the breaking down of heavy hydrocarbons into lighter fractions, the mixing of hydrocarbon molecules with oxygen molecules, and the chemical reaction within themselves, completing the combustion process. In order to make it possible for such combustion to occur in a small place, i.e., the combustion chamber, a high temperature, such as is provided by the combustion of an approximately stoichiometric mixture, is necessary. Since the stoichiometric ratio is about 15:1, and the overall Air-Fuel Ratios (AFR) at which Gas Turbine Combustors (GTC) operate at full power is between 33 and 40, it is necessary to introduce the air through three stages, which are called primary, secondary, and dilution zones. The primary zone initiates the combustion by mixing a portion of the air with fuel to create a high-temperature flame. The secondary zone introduces additional air to sustain combustion and ensure complete burning of the fuel. Finally, the dilution zone adds extra air to cool the exhaust gases, reducing their temperature before they enter the turbine, ensuring the safe operation of engine components downstream of the combustion chamber. This multi-zone design optimizes both combustion efficiency and temperature control within the chamber.

2.14. Types of Combustor

There are three main types of combustors for GTE's. These are the tubular, tubo-annular and annular. The selection of which type of combustor to use depends of the overall engine design and by the need to use the space as effective as possible.

2.15. Tubular Combustor

A tubular combustor, also known as a can combustor, is the simplest form of the combustor and was first used in early jet engines, such as the Whittle W2B and the Jumo 004. The major difference, however, between modern can combustors and the Whittle can combustor, is that this last had a reverse flow arrangement, as shown in figure 1.1, in order to permit a short shaft engine. This arrangement however, created a considerable pressure loss, thus the straight through can combustors where then developed by Joseph Lucas Limited.

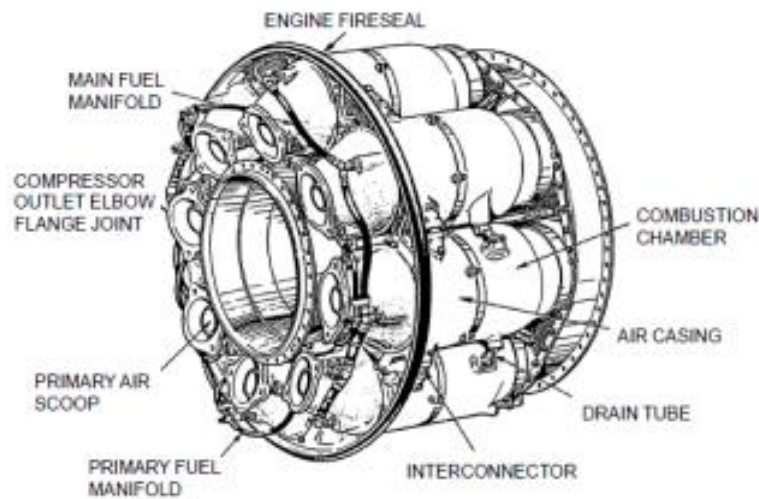


Figure 2.8. Tubular Combustor

A Gas Turbine Engine (GTE) can have multiple combustors around the engine, with the air supplied by the compressor directed through ducts into each can combustor. Each of these combustors comprises a cylindrical liner mounted concentrically inside an outer cylindrical casing, but they are all interconnected. This is to allow each of the combustors to operate at the same pressure and also allows combustion to propagate around the liners during engine starting. The interconnection of the combustors ensures uniform temperature distribution and consistent performance across the engine, reducing the risk of hotspots that could lead to damage.

2.16. Annular Combustor

The annular combustor consists of a single combustion chamber, which is annular in form, contained in an inner and outer casing. The chamber is opened at the front to the compressor and at the rear to the turbine nozzles. The combustor in study is of the annular type, and is illustrated in Figure 2.7.

With the introduction of the annular combustor came the fuel spray nozzle. This type of fuel injection greatly improves the preparation of fuel for combustion by aerating the over-rich pockets of fuel vapours that are close to the spray nozzle, thus big reductions in initial carbon formation are achieved.

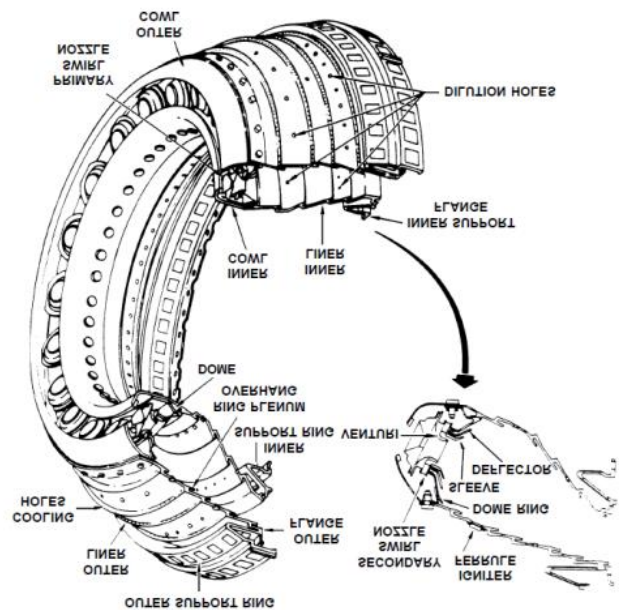


Figure 2.9. CFM56-3 Annular Combustor

2.17. Turbo-Annular Combustor

Due to the drawbacks of the tubular and annular combustors, the turbo-annular combustor was developed. These combustors are called so, or 'cannular', because they are basically a hybrid between the two basic types. They consist of a number of combustion chambers fitted in an annular casing. This arrangement combines the ease of overhaul and testing of the tubular type with the compactness of the annular type.

The turbo-annular also has a good match of fuel spray with airflow pattern and a low engine diameter, that can be obtained with the tubular and annular type, respectively. It is easy to develop and the rig testing is normally carried out using a segment containing a single combustion chamber. Tests including combustion

efficiency, flame stability, ignition limits, combustion chamber temperature and exhaust smoke, can be carried out. However, on segment rigs of this kind, it is not possible to reproduce the outlet temperature profile obtained with an actual engine.

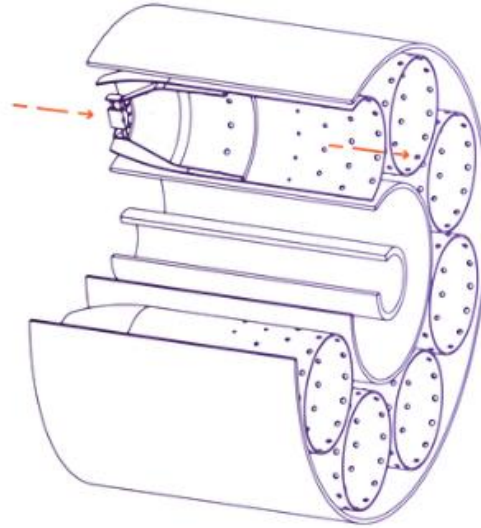


Figure 2.10. Turbo-Annular Combustor